

Counting Identities Nearest to the Dobbertin and Riverlane Patterns

A visual crosswalk through *Proofs That Really Count*

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Abstract

The closest links are not claims that Dobbertin’s finite-field theorem or the Riverlane experiments literally count Fibonacci tilings. They are shared *proof shapes*: partition into fibers, cut at an interface, telescope an orbit window, pair and cancel candidates, and propagate a local recurrence. This note identifies specific identities in Benjamin–Quinn’s *Proofs That Really Count*, states the precise algebraic analogues, and points to a machine-checked Lean module containing the exact reusable identities.

1 The shortest answer: six identities and techniques

Benjamin–Quinn	Core proof pattern	Nearest application here
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Identity 3, $f_{m+n} = f_m f_n + f_{m-1} f_{n-1}$	Cut at a boundary: breakable versus one tile crossing the seam	Riverlane A-windows versus reconciliation; local/global factorization
Identity 8 (Cassini), $f_n^2 = f_{n+1} f_{n-1} + (-1)^n$	Tail swap pairs almost everything; one exceptional configuration remains	Dobbertin’s two-candidate branch and its residual value $e_0 + e_1 = 1$
Identity 9, $\sum_{k=0}^n f_k^2 = f_n f_{n+1}$	Partition pairs by their last fault	Window decoding: classify a global history by its last unresolved boundary
Identity 129, Parity fixes one remaining binary choice $\sum_k \binom{2k}{n} = 2^{n-1}$	Dobbertin fixes one remaining binary choice	Dobbertin’s parity condition $k' + an \equiv 1 \pmod{2}$ as a branch-selection certificate
Identities 167–168, Sign-reversing pairing; unmatched alternating binomial boundary terms survive	Sign-reversing pairing; unmatched	Characteristic-two cancellation and the forbidden candidate pair
Chapter 3 linear recurrences	Condition on the final local tile and iterate	Frobenius orbit recurrence; repeated decoder/backlog rounds

The book’s overarching method—count one finite set in two ways, or exhibit a bijection—is even closer than any single numbered formula. For a map $f : A \rightarrow B$, the universal double-counting identity is

$$|A| = \sum_{b \in B} |f^{-1}(b)|. \tag{Fibre count}$$

Dobbertin proves that the generalized-Kasami evaluation is a permutation under its parity hypothesis. Consequently every output fiber has cardinality one:

$$|q_a^{-1}(b)| = 1 \quad (b \in \text{GF}(2^n)). \tag{Singleton fibers}$$

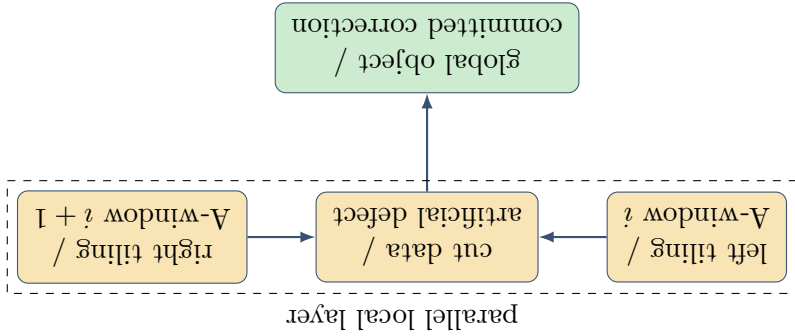
This is exactly “really counting”: the domain is partitioned by observed values, and the hard proof establishes that each part has one member.

2 Cut identities: Fibonacci seams and decoder boundaries

Benjamin–Quinn Identity 3 classifies an $(m+n)$ -tiling according to what happens at the cut after cell m . Either the tiling breaks there, contributing $f_m f_n$, or one domino crosses the cut, contributing $f_{m-1} f_{n-1}$. The exact Matlab indexing used in the Lean development is

$$F_{m+n+1} = F_m F_n + F_{m+1} F_{n+1}. \quad (\text{Fibonacci cut})$$

The structural parallel with Riviere’s 2023 parallel-window decoder is strong: local A-window results compose independently away from seams, while artificial defects at commit/buffer boundaries are exported to a B-window for reconciliation. Both arguments work because the global object is classified by a small *boundary interface*.



3 Telescoping: interiors cancel and endpoints remain

For streaming decoding, an analogous useful statistic is the last unresolved boundary or last round carrying pending work. The paper does not assert this Fibonacci formula; the correspondence is the exhaustive, disjoint partition by where the final obstruction occurs.

$$\sum_n^{k=0} f_n^2 = f_n f_{n+1}. \quad (\text{Last-fault partition})$$

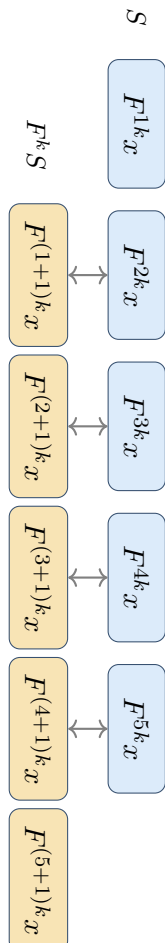
Identity 9 sharpens the seam idea by classifying a pair of tilings according to its *last fault*:

$$S(x) = \sum_{k'}^i x_{2^{ik}}.$$

The modular inverse relation $kk' \equiv 1 \pmod{n}$ closes the cyclic endpoint. The exact identity is

$$S(x)^{2^k} = S(x) + x_{2^k} + x_{2^{k+1}}. \quad (\text{Dobbertin endpoint shift})$$

In characteristic two, duplicated interior terms cancel. This is the same visual grammar as a telescoping counting argument: shift the indexed family, pair the common interior, and retain only boundary terms.



common interior cancels; cyclic congruence identifies the far endpoint

This is closest to the book's many "condition on the last tile/fault" and alternating-cancellation proofs, rather than to a numerical Fibonacci identity. The invariant version is the trace identity

$$\mathrm{Tr}(x^{2^k}) = \mathrm{Tr}(x),$$

which says a cyclic shift changes no total orbit count/sum.

4 Pairing, parity, and one exceptional object

Cassini's Identity 8 is proved by tail swapping. Faulty tiling pairs match; exactly one fault-free pair remains, with its side determined by parity:

$$f_n^2 - f_{n+1}f_{n-1} = (-1)^n.$$

The Dobbertin gamma branch has a notably similar architecture. Two distinct candidates in a residual fiber are forced into a pair with

$$x_0 + x_1 = \Delta, \quad \mathrm{Tr}(x_0) = \mathrm{Tr}(x_1), \quad e_0 + e_1 = 1. \quad (\text{Candidate-pair identity})$$

But two candidates satisfying the original equation would require equal residual labels. In characteristic two, equality gives $e_0 + e_1 = 0$, contrary to the displayed 1. Thus the pair is forbidden. This is an "almost-bijection plus exceptional term" proof in the precise sense familiar from Cassini and the book's signed involutions.

Identity 129 gives the cleanest parity analogy:

$$\sum_k \binom{n}{2k} = 2^{n-1}.$$

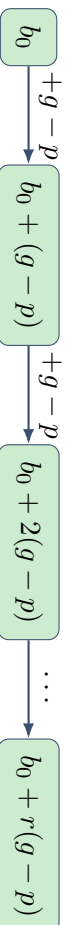
After freely choosing $n - 1$ membership bits, the last bit is forced by desired parity. In Dobbertin, $k' + an \equiv 1 \pmod{2}$ similarly determines which parity branch is admissible. The analogy is structural only: it does not turn the field equation into a committee-counting problem.

5 Iteration and the Riverlane rate identity

Chapter 3 treats a local recurrence by conditioning on the final tile and then iterating. Riverlane's throughput abstraction has the first-order affine recurrence

$$b_{r+1} = b_r + g - p, \quad b_r = b_0 + r(g - p). \quad (\text{Backlog identity})$$

Therefore $g \leq p$ makes the upper-bound invariant propagate through every round. This exact closed form separates two paper-level notions: throughput is the slope $g - p$, while latency concerns the delay along the composed processing path. The 2023 paper obtains throughput through parallel windows; the 2026 paper demonstrates an integrated low-latency FPGAs path and discusses streaming operation. The identity is a deterministic abstraction, not a formalization of their experimental timing data.



6 What is machine checked

The module `RequestProject/StandaloneDobbertin/Structure/CountingIdentities.lean` proves:

- the finite fiber double-counting formula and singleton-fiber consequence of bijectivity;
- the exact all-round backlog formula;
- the characteristic-two impossibility of equal labels summing to one;
- the Fibonacci cut identity;
- Dobbertin’s exact Frobenius endpoint-cancellation identity.

These connect to the already formalized generalized-Kasami permutation theorem, candidate-pair identity, Riverlane boundary interface, and no-backlog invariant. No assertion is made that the Riverlane papers’ physical, statistical, or performance claims follow from these counting identities.

Sources. Arthur T. Benjamin and Jennifer J. Quinn, *Proofs That Really Count: The Art of Combinatorial Proof* (2003), especially Identities 3, 8, 9, 129, 167–168 and Chapter 3. Hans Dobbertin, “Kasami Power Functions, Permutation Polynomials and Cyclic Difference Sets,” Theorem 1. L. Skoric et al., “Parallel window decoding enables scalable fault tolerant quantum computation” (2023). L. Caune et al., “Demonstrating real-time and low-latency quantum error correction with superconducting qubits” (2026).